Accident Detection Using YOLO and Two-Stream CNN

Presented by
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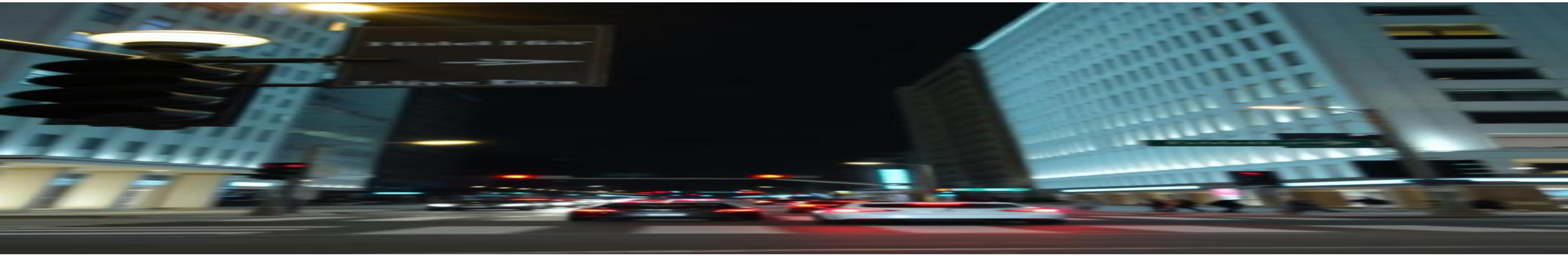


Brief Overview

Problem
Traffic accidents are a leading cause of fatalities globally, and current detection systems struggle with real-time processing, accuracy, and scalability.

Solution
Proposes a dual-methodology approach using YOLOv8 for static image datasets and Two-Stream CNN for dynamic video datasets.

Introduction

Road Accidents

Road accidents are a major challenge, leading to millions of deaths annually.

Existing Systems Limitations

Current systems face limitations with real-time processing, dataset scalability, and detection accuracy.

Proposed Solution

Aims to address these challenges by developing advanced computer vision techniques combining YOLO and Two-Stream CNN for smart city traffic safety initiatives.

Objectives

YOLO

Develop a YOLO-based system for detecting accidents in static image datasets.

Two-Stream CNN

Design and implement a Two-Stream CNN framework for video-based accident detection.

Evaluation

Evaluate the performance of the proposed methods against existing techniques.

Scalability

Create a scalable and deployable solution for future real-world traffic monitoring systems.

Literature Review

Title	Key Focus	Boundaries and Limitations
Attention R-CNN for Accident Detection	Considering variations in light intensity.	Lacks detailed exploration of real-world implementations and broader performance validation
Two-Stream CNN for Action Recognition in Videos	Two-Stream CNN for action recognition	Resource-intensive; unsuitable for lightweight, real-time systems.
Accident Detection Using Deep Learning	LSTM and CNN algorithm	Focuses more on concept than detailed implementation or validation.
A Novel Dataset for CCTV Traffic Camera based Accident Analysis	LSTM and RCNN algorithm	Dataset limited to YouTube videos.

Methodology

Data Collection

Find also create static images and video datasets representing diverse traffic conditions.

Data Preprocessing

Resize, normalize, and annotate images and videos for training.

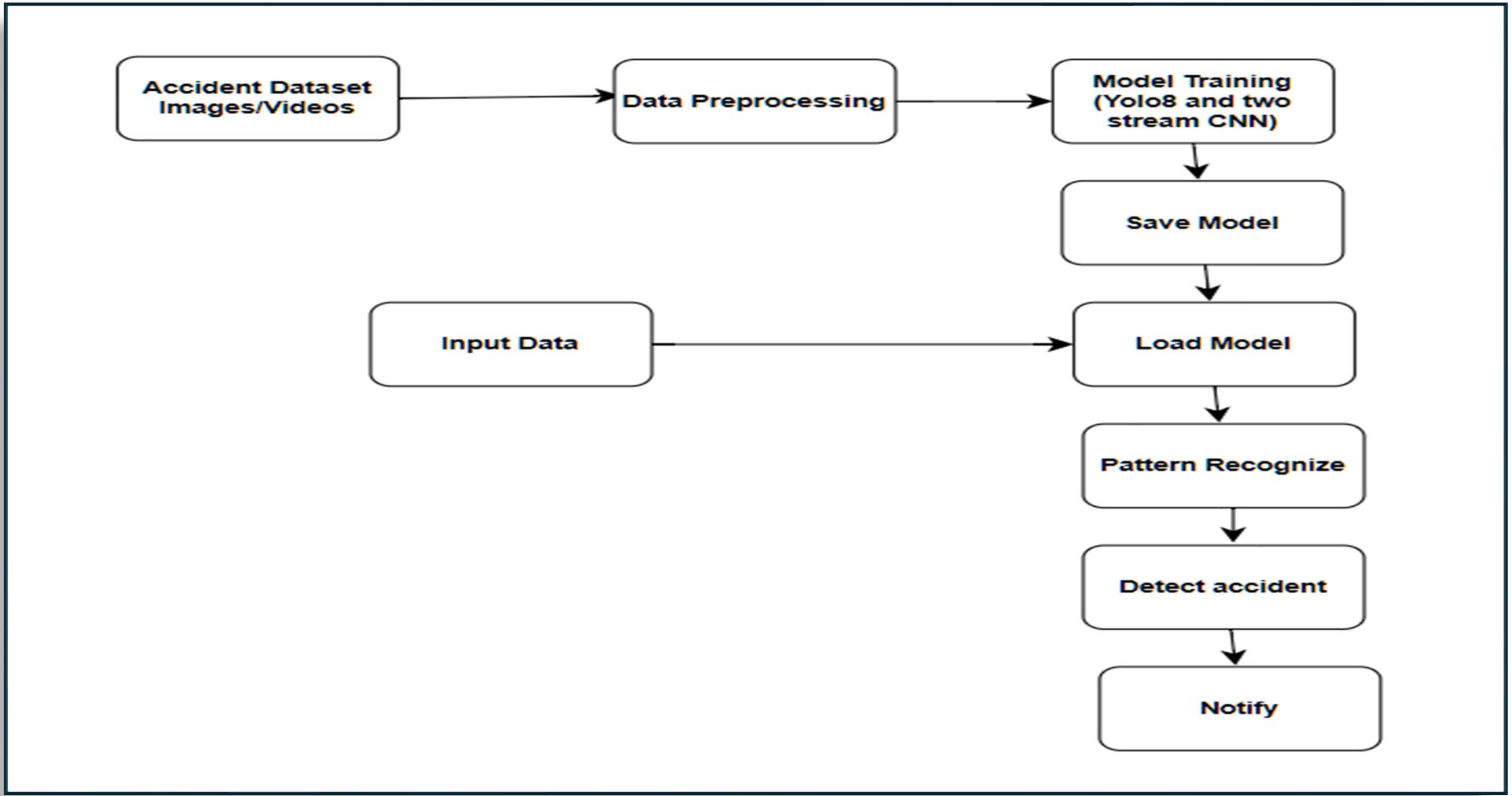
Model Training

Train YOLOv8 on image datasets and Two-Stream CNN on video datasets.

Evaluation

Assess model performance using metrics like mAP, F1-score, and Precision.

Methodology



Expected Outcomes and Contributions

1

YOLO Model

A YOLOv8-based model for detecting accidents in static images.

2

Two-Stream CNN Model

A Two-Stream CNN model for detecting accidents in videos.

3

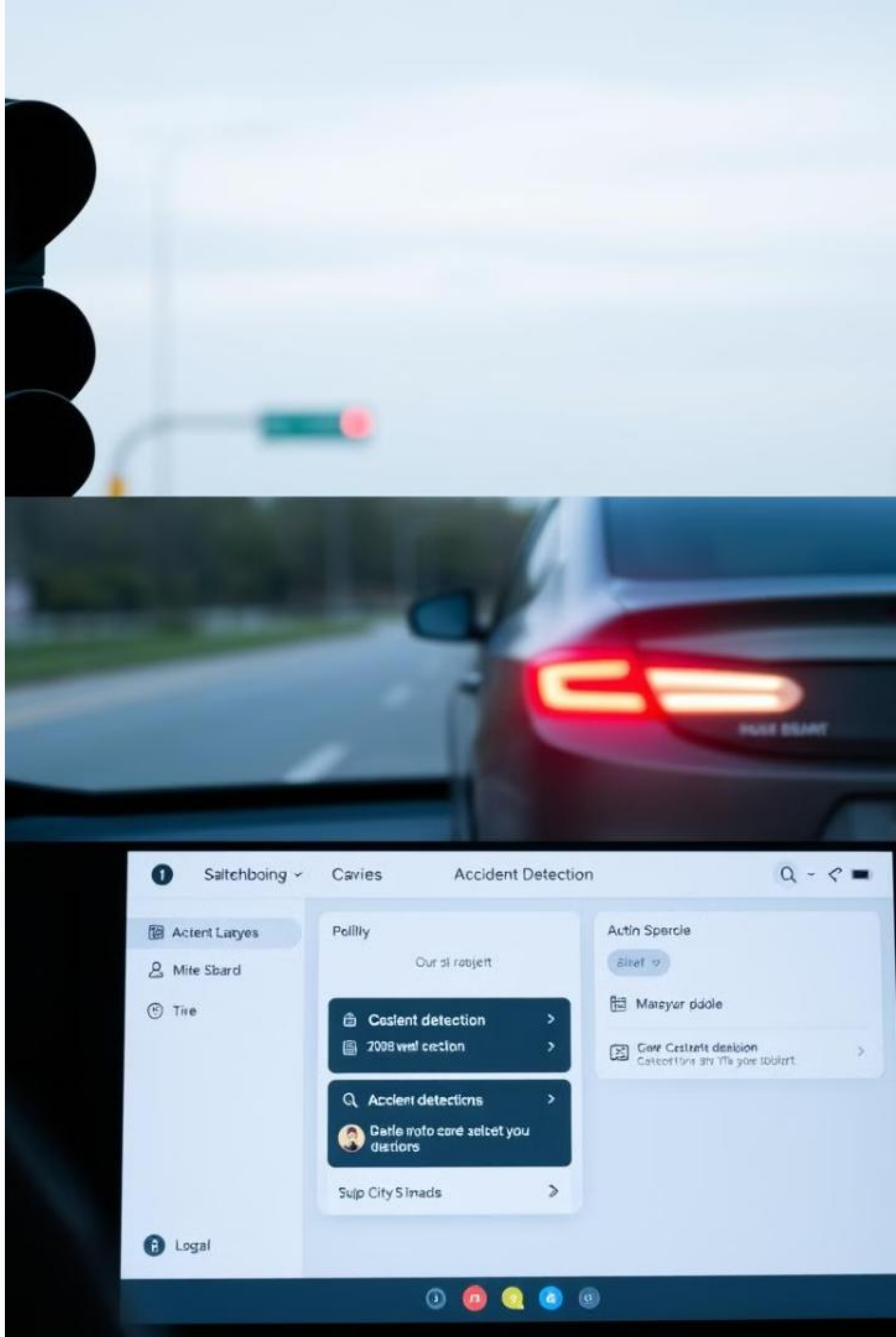
Improved System

A system for improved accident detection accuracy and scalability.

4

Future Research

Insights for future research in traffic safety systems.



Conclusion

1

Challenges Addressed

Addresses critical challenges in accident detection by leveraging YOLOv8 and Two-Stream CNN.

2

Enhanced Safety

Aims to significantly enhance road safety and emergency response capabilities.

3

Societal Impact

Approval and support for drive advancements in intelligent transportation systems, benefiting both research and societal needs.

References

- Ankit Shah " CADP: A Novel Dataset for CCTV Traffic Camera based Accident Analysis"
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